

Developing a Composite Hydrogel Brain Model as a tool for Traumatic Brain Injury Research

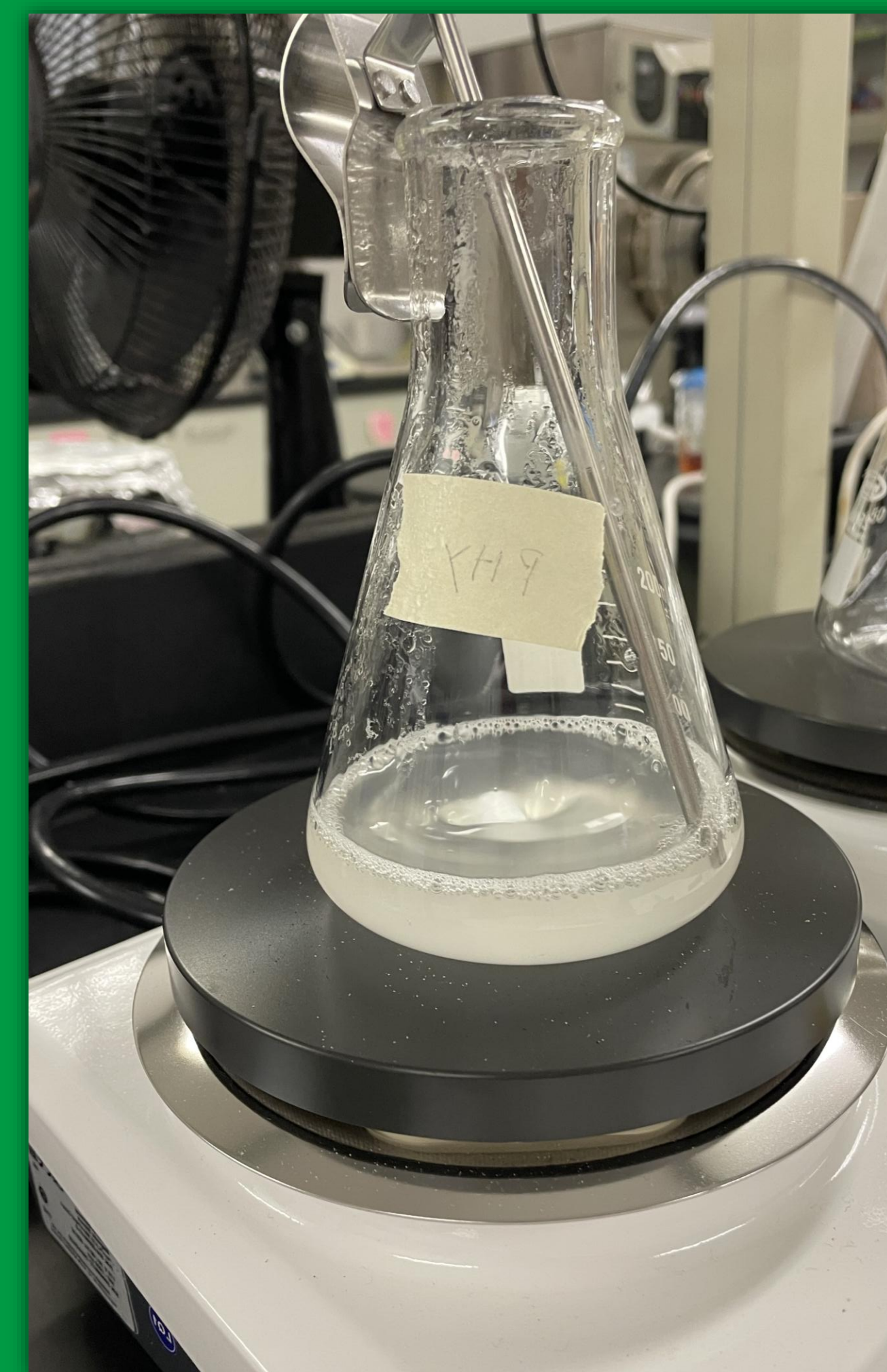
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Introduction

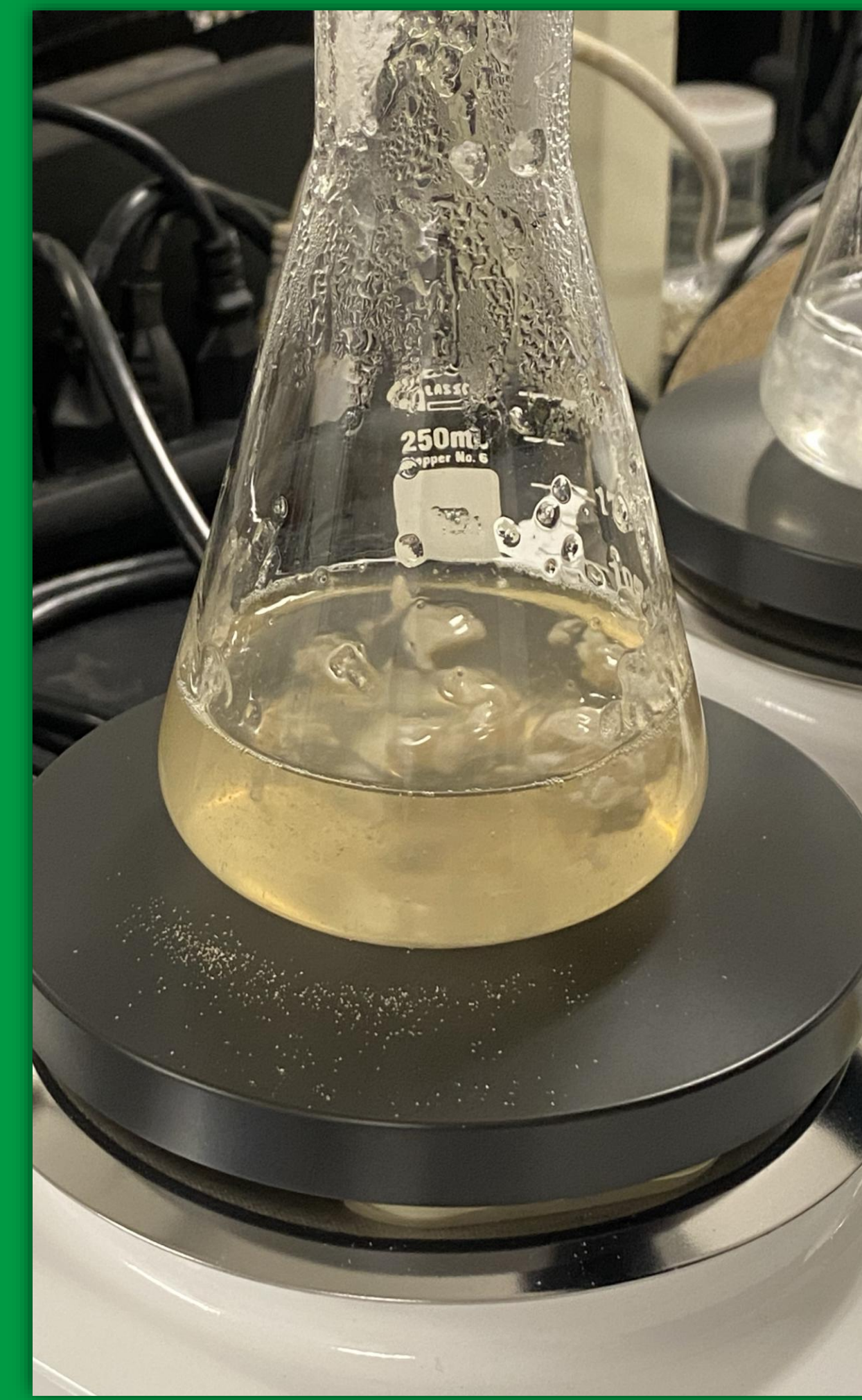
Traumatic Brain Injury (TBI) remains a major public health concern, with over 200,000 hospitalizations and 70,000 deaths annually, yet ethical and practical limitations hinder direct study of human brain tissue under high-impact conditions. This study presents a novel biofidelic brain model using composite hydrogels engineered to replicate brain tissue mechanics at high strain rates, specifically the Young's Modulus values for the cortex as described by Budday et al. (2017) with a neo-Hookean model of $E = 5.9$ KPa, capable of enabling more accurate investigation of TBI mechanisms and supporting the development of improved prevention and recovery strategies.

Methodology

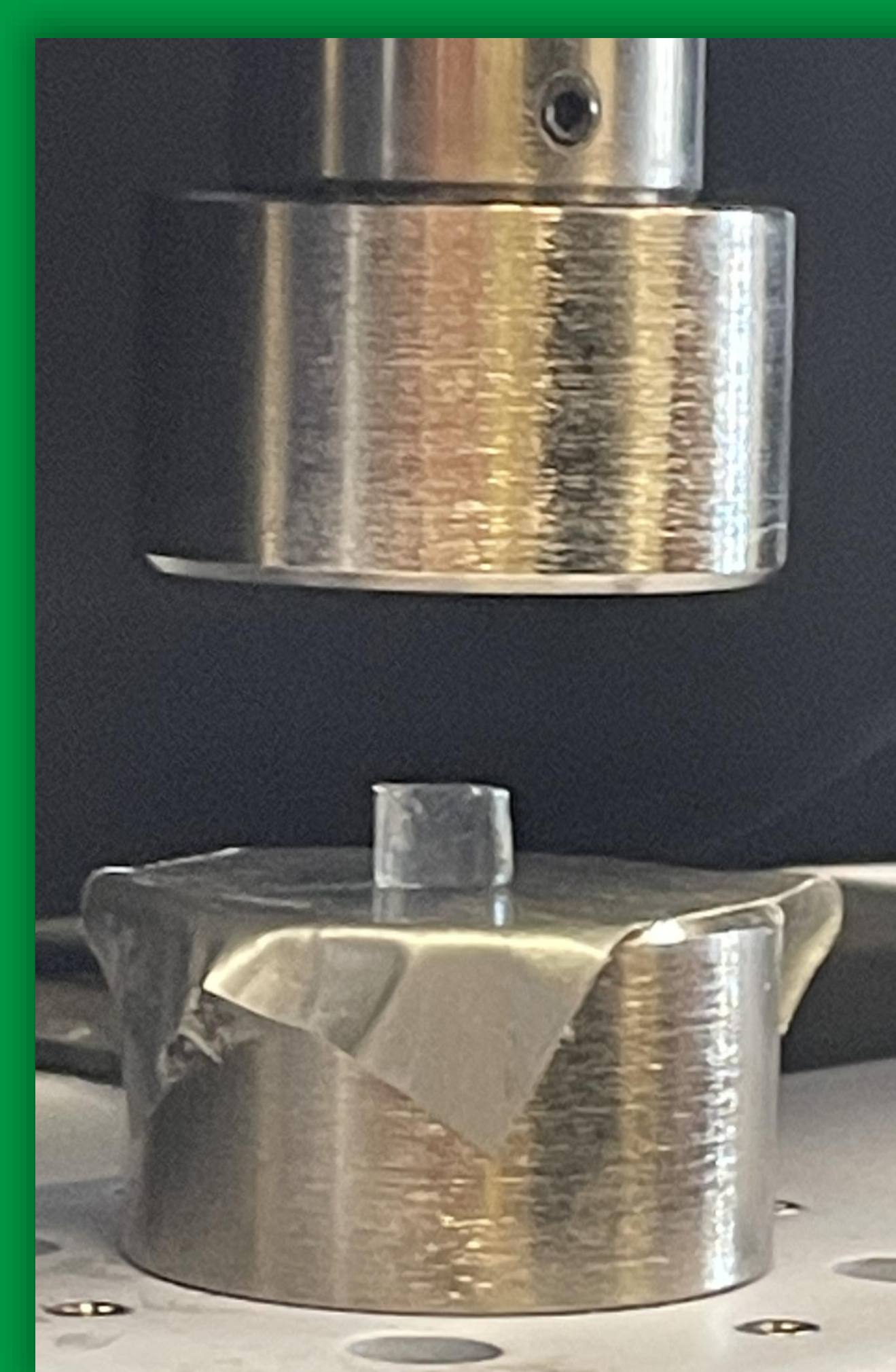
To develop the model, a variety of composite hydrogels were formulated and evaluated for their ability to replicate the mechanical properties of human brain tissue under high strain rates in accordance with Budday et al. (2017) mechanical characterization. These included a low Young's Modulus and nonlinear viscoelastic behavior. The primary hydrogel component, polyvinyl alcohol, was combined with additional biocompatible polymers to alter the hydrogel material properties. The hydrogels were prepared using deionized water and subjected to 3-5 freeze-thaw cycles to reinforce physical crosslinking of the polymers. The resulting hydrogels were molded into petri dishes, and 10 mm biopsy punches were used to create cylindrical samples of ~8 mm height and 10 mm diameter, with which preliminary evaluation in compression was performed. Isolated materials, such as Phytigel components alone, even at lower concentrations, gave us much stiffer and less compliant hydrogels than current mechanical characterization of the brain at $E = 5.9$ KPa and our composites we developed. (9.2 kPa vs 11.6 kPa)



(a). PVA During Synthesis



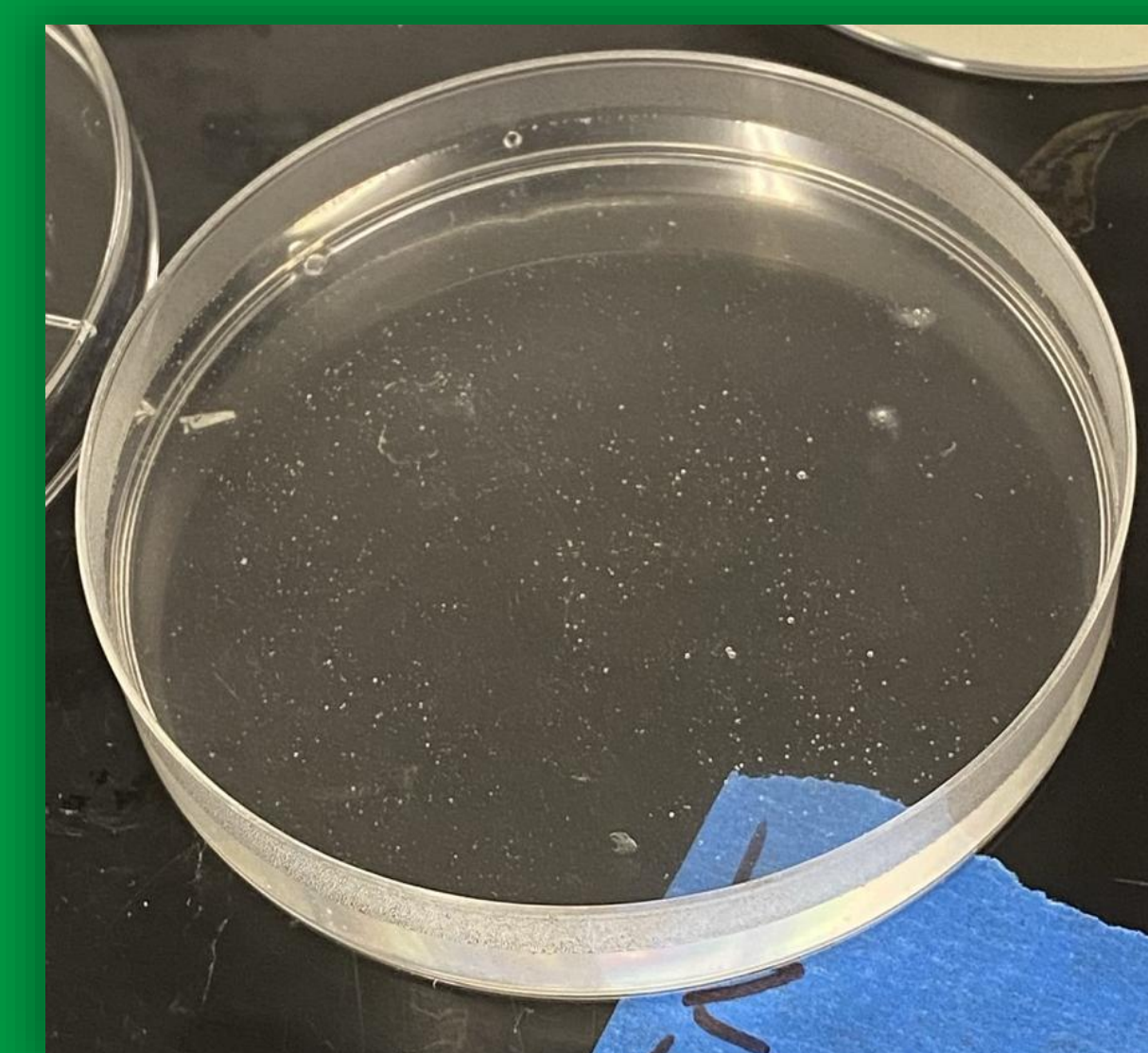
(b). PHY During Synthesis



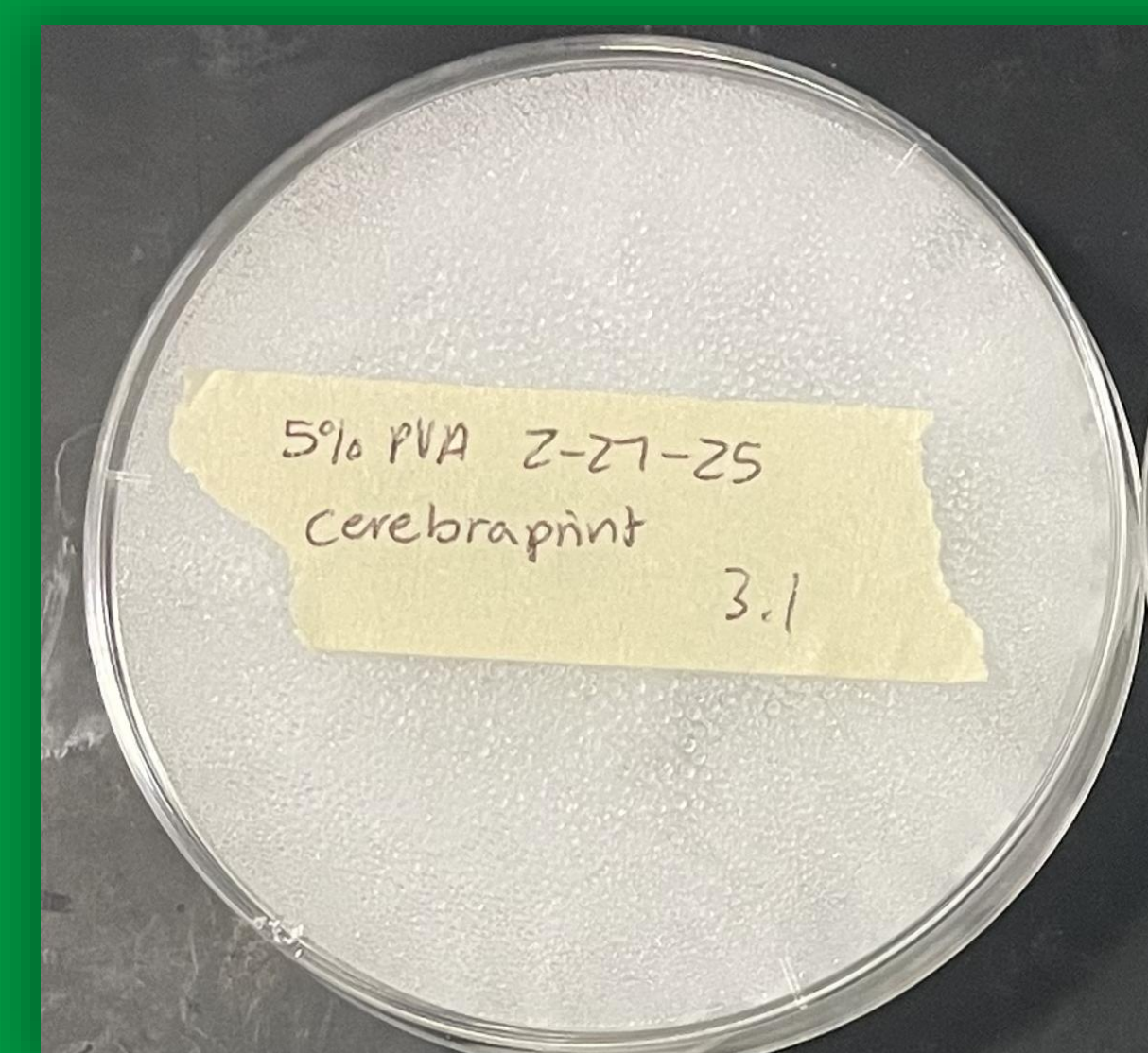
(e). Before Compression



(f). During Compression



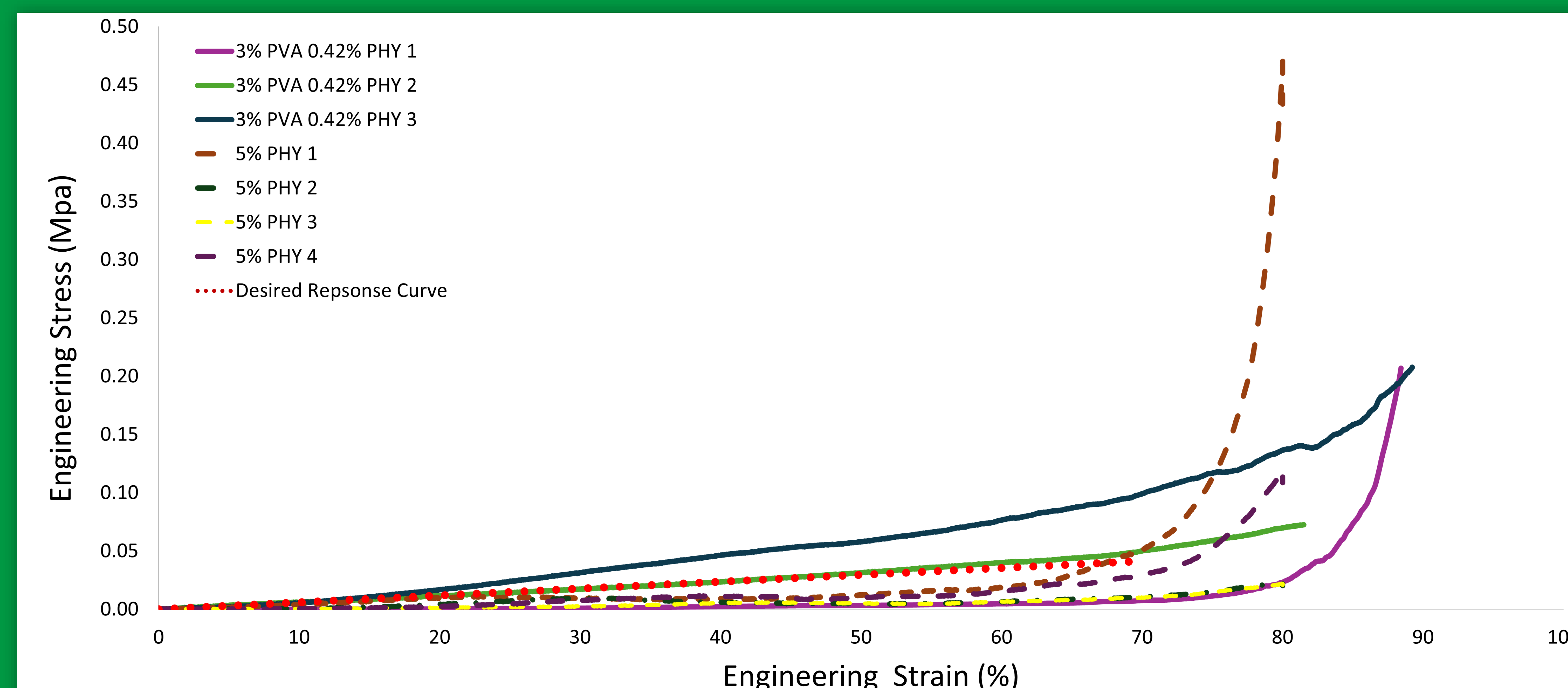
(c.) Before Freeze cycle



(d.) After Freeze Cycle

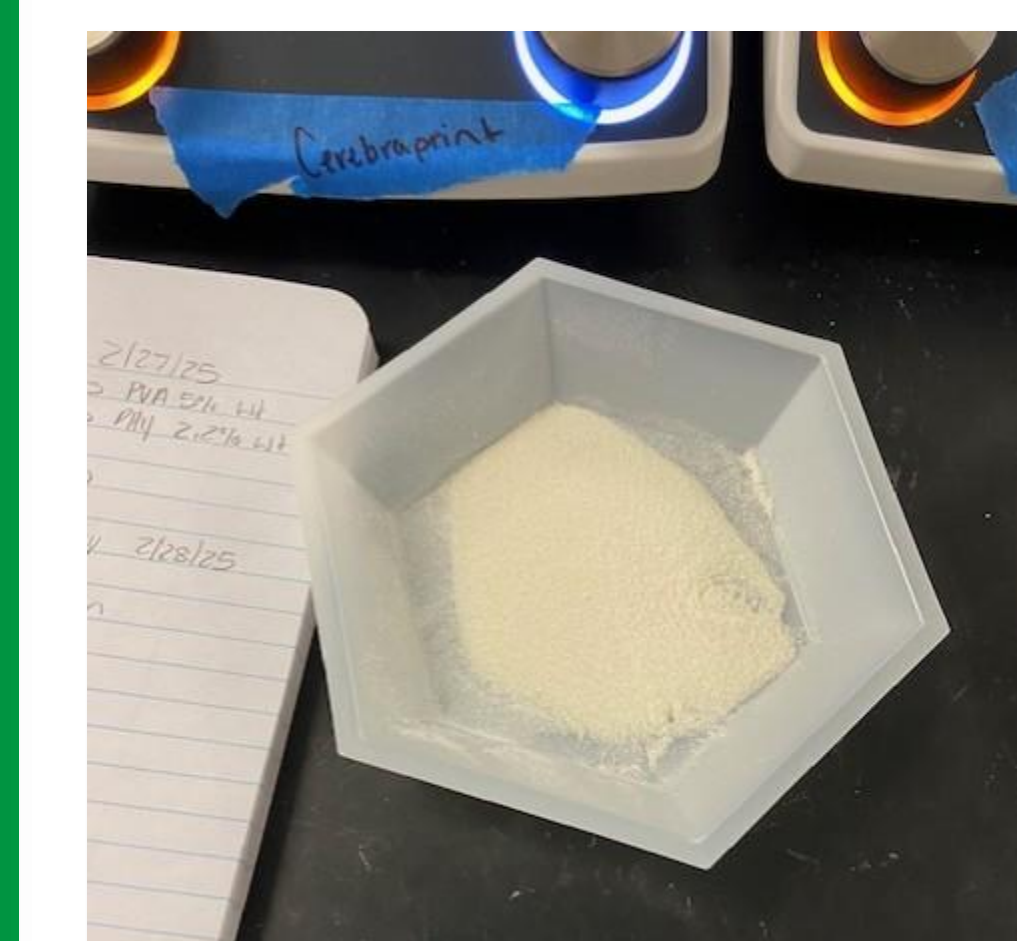


(g) Hydrogel Sample From Biopsy Punch (Gloved Hand)



Conclusion

The stress-strain response of our 3% PVA / 0.425% PHY composite samples (5-7) reveals nonlinear, strain-dependent mechanical behavior consistent with human brain tissue, displaying Young's Modulus Values of 9.2 kPa, which is stiffer than the cortex tissue characterized by compressive tests in Budday et al. (2017). This shows a discrepancy of 3.3 kPa, concluding more work needs to be done. All samples exhibited low initial stiffness and gradual strain stiffening beginning near 15–20% strain. While slight differences were observed across the four samples, all were fabricated using the same homogeneous formulation. These differences likely reflect minor experimental factors such as casting imperfections, microbubbles, or test alignment, rather than material heterogeneity. Compared to hydrogels with 5% PHY, (11.6 kPa), this formulation balances compliance and mechanical stability, making it a potential first step for brain tissue modeling — particularly grey matter, the cortex portion of the brain. Future samples will vary in combinations of PHY and PVA to vary the stiffness of the material as well as gather more information about how each percentage change impacts the Young's modulus at high strain rates.



(h). 2.23 g PHY



(i). 5.01 g PVA

For more information scan here

